

Day 19 - Lecture 19.14: Distribution Plots in Seaborn (`displot`, `histplot`, `kdeplot`)

1. Overview & Purpose

Distribution Plots allow us to inspect the shape, spread, modality, and skewness of continuous variables. Modern Seaborn (v0.11+) provides:

- `sns.histplot()` : Modern, powerful replacement for `plt.hist()`.
 - `sns.kdeplot()` : Kernel Density Estimation (smooth probability density curve).
 - `sns.displot()` : Figure-level function providing access to histograms, KDEs, and ECDFs with automated subplot faceting.
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2. Kernel Density Estimation (KDE) Theory

A histogram's appearance depends heavily on arbitrary bin boundaries. KDE replaces discrete bins with a smooth, continuous estimate of the probability density function $\hat{f}(x)$:
$$\hat{f}(x) = \frac{1}{n h} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right)$$
 where $K(\cdot)$ is a Gaussian kernel function and h is the smoothing bandwidth.

- In `sns.histplot()`, setting `kde=True` overlays this smooth mathematical density over the discrete bins.
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3. Code Implementation: The Penguins Dataset

```
import seaborn as sns
import matplotlib.pyplot as plt

penguins = sns.load_dataset("penguins")
print("Penguins Dataset Preview:")
print(penguins.head())

# 1. Distribution of Body Mass (30 bins with KDE overlay)
plt.figure(figsize=(9, 5))
sns.histplot(
    data=penguins,
    x="body_mass_g",
    bins=30,
    kde=True,
    color="teal"
)
plt.title("Penguin Body Mass Distribution with KDE Overlay", fontsize=14, fontwei
plt.xlabel("Body Mass (grams)", fontsize=12)
plt.ylabel("Count", fontsize=12)
plt.grid(axis='y', linestyle='--', alpha=0.6)
plt.show()

# 2. Multi-class distribution with hue
plt.figure(figsize=(9, 5))
sns.histplot(
    data=penguins,
    x="body_mass_g",
    hue="species",
    element="step", # 'bars', 'step', or 'poly'
    kde=True
)
plt.title("Body Mass Distribution Across Penguin Species", fontsize=14, fontweigh
plt.xlabel("Body Mass (grams)")
plt.show()
```

4. Key Takeaways

- `sns.histplot()` handles missing values (`NaN`) gracefully without crashing.
- Setting `element="step"` or `multiple="stack"` creates clean, uncluttered visual comparisons across multiple species.

